

INFORMATION AND RANDOMNESS

MIDSEMESTER EXAMINATION

COURSE CODE: CS-510

FULL MARKS: 50

Attempt All Questions

Time: 2 hours

1. Given a periodic sequence $s = 011000111$ of period length $N = 9$ determine the number of **blocks** and **gaps** and the value of **autocorrelation** coefficients for $C(t)$ $0 \leq t \leq 8$?

1 + 1 + 4 = 6

2. In this exercise, we consider some properties of **Linear Congruential Generator** defined by $s_i = (as_{i-1} + b) \bmod M$. All operations are performed in Z_M and we assume that $a \neq 1$.

(a) Prove that $s_i = s_0 a^i + \frac{b(a^i - 1)}{a - 1}$ for all $i \geq 0$.

(b) The period of a **Linear Congruential Generator** is the smallest positive integer t such that $z_{i+t} = z_i$. For all $i \geq 0$. Prove that $t = 1$ if $s_0 = b/(a-1)$

(c) Prove that the period of t satisfies the equality $t \leq n$, where n is the order of a in Z_M^* .

3 + 5 + 2 = 10

3. Consider the linear congruential generator defined by $s_i = (3s_{i-1} + 1) \bmod 19$. Show that the next bit predictor $\mathbf{nbp}(z) = 1 - z$ is an ϵ -next bit predictor for the i^{th} bit and determine the value of ϵ . Justify your answer.

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4. In the class we proved that, in order to reduce the error probability of an unbiased Monte Carlo algorithm from $\frac{1}{2} - \delta$ to γ , where $\gamma + \delta < \frac{1}{2}$, it suffices to run the algorithm m times, where

$$m = \left\lceil \frac{1 + \log_2 \gamma}{\log_2 (1 - 4\delta^2)} \right\rceil$$

Prove that this value of m is $O(1/\gamma\delta^2)$.

5. If X is a normally distributed random variable having mean 4 and variance 25 what should be the threshold values of x_α to achieve the significance level of $\alpha = 0.01$ given the following table of standard normal distribution:

α	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
x	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902	3.2905

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6. Suppose we build an LFSR machine that works mod 3 instead of mod 2. It uses a recurrence of length 2 of the form

$$x_{n+2} = c_0 x_n + c_1 x_{n+1} \pmod{3}$$

to generate the sequence 1, 1, 0, 2, 2, 0, 1, 1. Set up and solve the matrix equations find the coefficients c_0 and c_1 .

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7. Suppose you modify the LFSR method to work mod 5 and you use a (not quite linear) recurrence relation.

$$x_{n+1} = c_0 x_n + c_1 x_{n+1} + 2 \pmod{5}$$

$$x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 0$$

Find the coefficients c_0 and c_1 .

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